Scalability & Future Plan

Date	24October 2024
Team ID	
Project Name	Global Olympic Sports Performance Analysis
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Static Data Source (CSV)	Cannot update automatically for the 2028 Olympics.	High
2	Local Deployment	Users must run python app.py locally to view the portal.	High
3	Tableau Public Limits	Data is public and cannot be secured behind login walls.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Local server (Waitress/Werkzeug)	Deploy to Google Cloud Run or Heroku with load balancing.
2	Data Storage	Manual CSV upload to Snowflake	Implement Snowflake Snowpipe for automated data ingestion via AWS S3.
3	Performance	Standard Tableau rendering	Implement Tableau Data Extract optimizations and Webhooks.
4	Security	No authentication	Integrate Auth0 into Flask for Role-Based Access Control (RBAC).

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline
Phase 2	Cloud Deployment & Live API Integration (IOC Data)	Q1 2027
Phase 3	Predictive Machine Learning Models for 2028 Olympics	Q2 2027
Phase 4	Mobile Application (React Native) for Dashboard viewing	Q4 2027